

Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

Chapter 4: The Macroeconomy in the Short Run

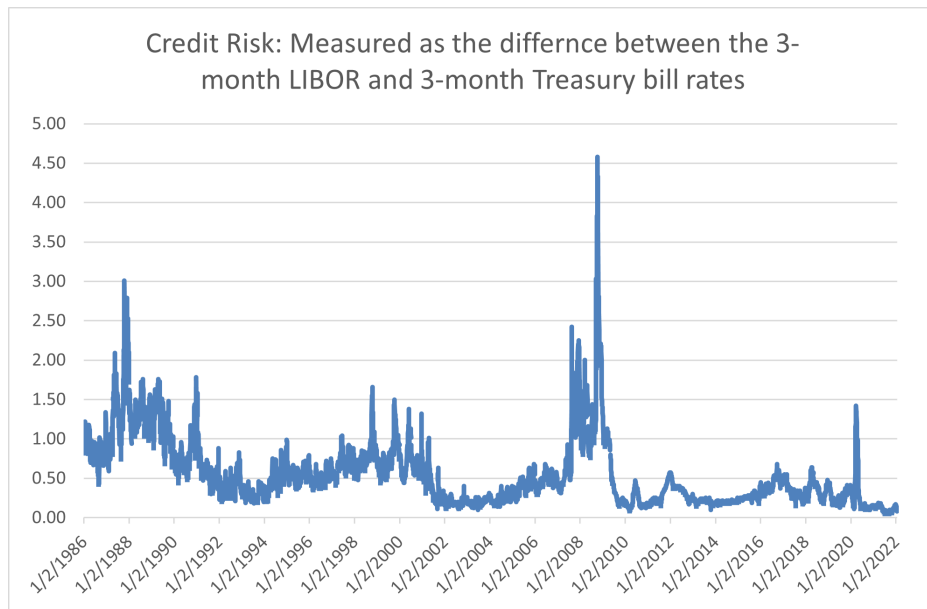
Dr. Le Van Ha

Vietnamese-German University

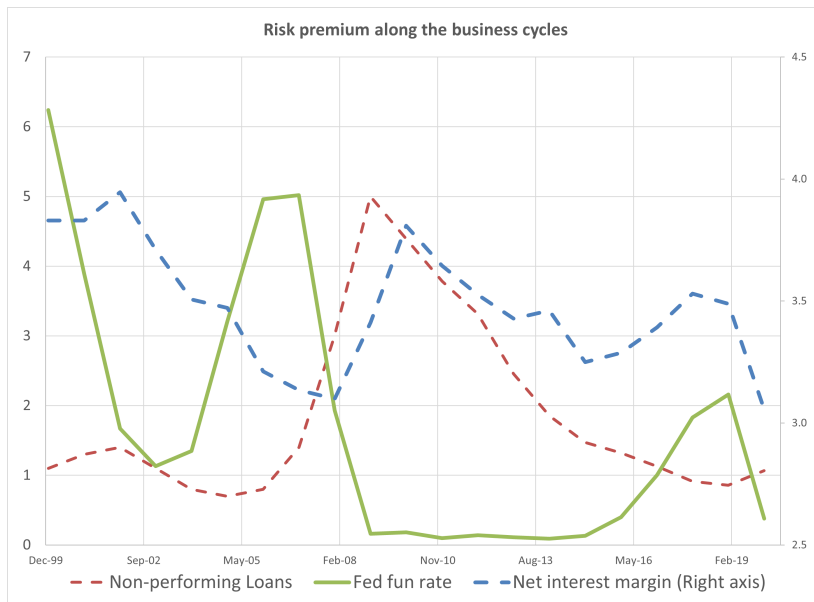
May, 2026

- + In the fall of 2008, the global financial crisis started with the collapse of Lehman Brothers, creating turmoil in financial markets, and leading to the chairman of the U.S. Federal Reserve stating during a weekend in late September 2008:
"If we don't do this [the Troubled Asset Relief Program], we may not have an economy on Monday."
 - + Preceding these events was a collapse of U.S. housing prices that was unprecedented in magnitude:
 - + Factors driving the U.S. housing prices in the early 2000's:
 - Global saving glut both in emerging market and industrial economies;
 - Housing becoming available to subprime borrowers (borrowers with poor credit records and/or high debt-to-income ratios), in part due to combination of low interest rates and questionable lending standards;
 - Following the Taylor rule, the U.S. Federal Reserve from about 2004 onwards started raising interest rates, which resulted in the default of many borrowers with adjustable rate mortgages (in August 2007: 16% of subprime mortgages with adjustable rates);
 - A sizable fraction of subprime mortgage loans had been lumped together by financial markets through securitization: mortgage-backed securities or collateralized debt obligations (CDOs);
 - Large share of these CDOs held by commercial and investment banks; following the collapse of the value of these CDOs, these banks ran into solvency problems;
- "Agency [bank management knowingly gambling, expecting a bailout] and neglected risk are the two most credible explanations of the financial crisis, but neglected risk is most parsimonious, consistent with ratings, expectations data, and regulator passivity in 2007. Combination of agency and neglect of risk would lead to massive risk taking by investors failing to appreciate exposure. This probably is the real story."*





Let's discuss the pattern of these three series during the business cycle.



- + We reconsider the **risk premium**—the mark-up commercial banks are charging households and firms to borrow from them.
- + We will augment our model of the macroeconomy in the short run to account for the fact that the value of loan collateral may drop sizably in crisis times (as did housing prices in the U.S.), in turn sizably increasing the lending risk (increasing the uncertain component of loan repayments), rendering the **risk premium endogenous** and further propagating initial shocks to the risk premium.
- + A second feature to consider in augmenting our model of the macroeconomy in the short run: In responding to the global financial crisis, the U.S. Federal Reserve
 - quickly lowered its monetary policy rate, the Federal Funds Rate,
 - but did not lower it below zero:
the so-called **zero-lower bound** for (nominal) interest rates that capture costs of borrowing.
- + Relative to the *IS-TR* model, the first novel component of what we will label the **extended IS-TR Model** is that we capture the zero lower bound on R^{MP} .
- + This zero lower bound can be understood as a no-arbitrage condition: If a commercial bank at a negative value of R^{MP} was paid interest to borrow, it could earn ever higher profits simply by ever increasing its amount of borrowing (presuming for simplicity the absence of any cost for storing the funds borrowed).

- + As in the macroeconomy in the short run prices are fixed, recall that we have that

$$\pi = \pi^T = \pi^e = 0,$$

implying

$$r^{MP} = R^{MP},$$

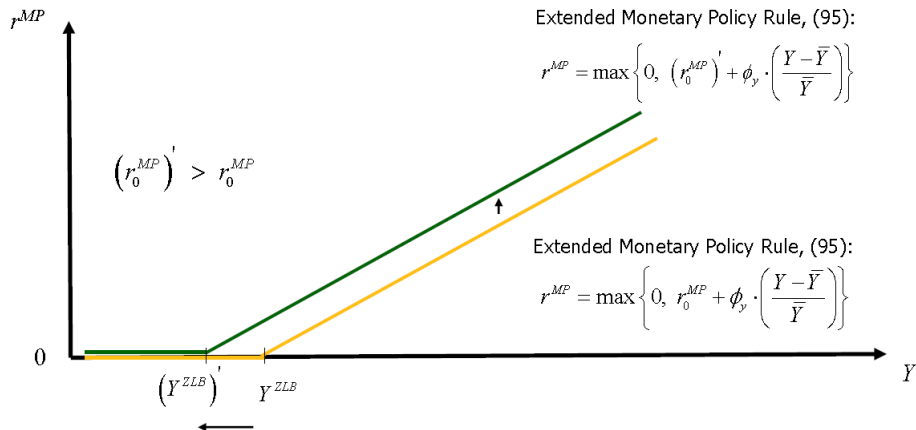
and we can thus capture the zero lower bound with the following **extended monetary policy rule**:

$$r^{MP} = \max \left\{ 0, \underbrace{r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right)}_{\text{Taylor Rule}} \right\}. \quad (95)$$

- + Under (95), if output is so low that according to the Taylor rule the central bank would choose to set a negative r^{MP} , it is constrained by the zero lower bound and instead sets $r^{MP} = 0$.
- + We will henceforth denote the highest level of output at which the central bank, following the Taylor rule, would set $r^{MP} = 0$ as " Y^{ZLB} ".
- + Whenever output is at or below Y^{ZLB} , we will call the macroeconomy to be in a **"deep crisis"**.
- + Note that Y^{ZLB} can be readily obtained from the extended monetary policy rule (95):

$$\begin{aligned} r_0^{MP} + \phi_y \cdot \left(\frac{Y^{ZLB} - \bar{Y}}{\bar{Y}} \right) &= 0 \\ \Leftrightarrow Y^{ZLB} &= \frac{\bar{Y}}{\phi_y} \cdot (\phi_y - r_0^{MP}). \end{aligned} \quad (96)$$

Y^{ZLB} depends on the parameters and variables entering the Taylor rule in economically plausible form: For example, if the neutral real rate r_0^{MP} was to increase, then the central bank sets the monetary policy rate to be higher than before (as long as the new zero lower bound constraint is not binding). This in turn implies that the central bank reaches the zero lower bound for r^{MP} at a lower level of output.



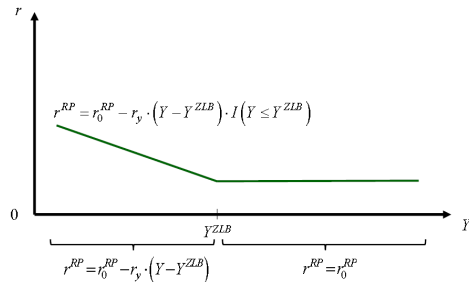
- + We capture the **endogenous adjustment of the risk premium** in "deep crisis" times through the following specification:

$$r^{RP} = r_0^{RP} - \underbrace{r_y \cdot (Y - Y^{ZLB}) \cdot I(Y \leq Y^{ZLB})}_{\substack{>0 \\ \text{endogenous component of risk premium}}}, \quad (97)$$

where I denotes the indicator function:

$$I(Y \leq Y^{ZLB}) = \begin{cases} 1 & \text{if } Y \leq Y^{ZLB} \\ 0 & \text{if } Y > Y^{ZLB} \end{cases} \quad (98)$$

Note that the specification in Equation (97) implies that within deep crises, the endogenous component of the risk premium rises by r_y for any unit decrease of output. The following graph illustrates this:



Taking into account both

- the zero lower bound on the monetary policy rate, and

- the endogenous component of the risk premium in "deep crisis" times,

we now have the following financial sector real interest rate function (the **extended TR-curve**):

$$r = \underbrace{\max \left\{ 0, r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) \right\}}_{=r^{MP}} + \underbrace{r_0^{RP} - r_y \cdot (Y - Y^{ZLB}) \cdot I(Y \leq Y^{ZLB})}_{=r^{RP}}. \quad (99)$$

The extended *TR*-curve features a discontinuity at that level of output for which the Taylor rule suggests switching from a positive to a zero monetary policy rate, Y^{ZLB} .

When $Y > Y^{ZLB}$, the extended *TR*-curve (99) reduces to

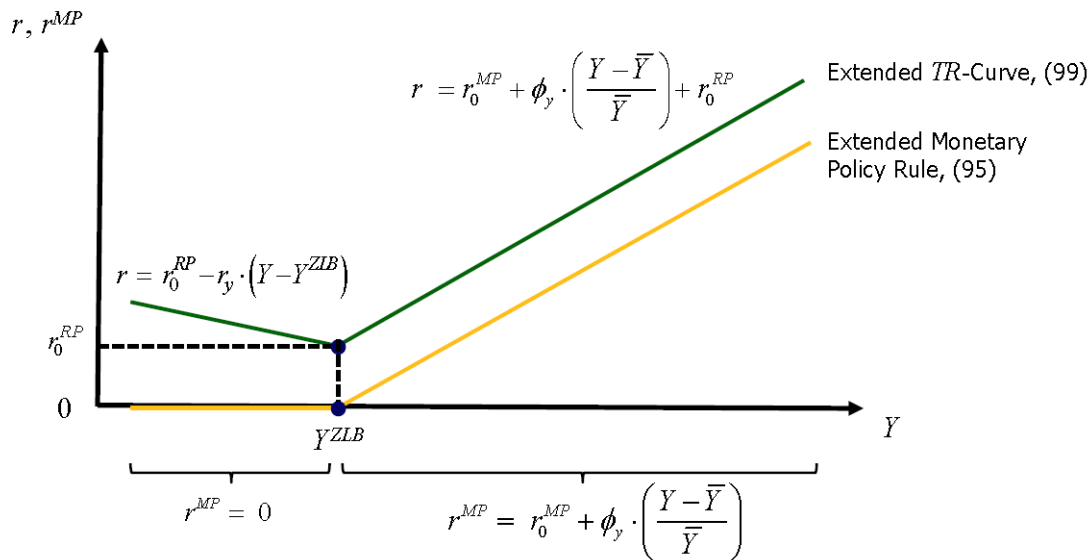
$$r = r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) + r_0^{RP},$$

the standard *TR*-curve given by (75): the interest rate at which households and firms can borrow is the sum of the monetary policy rate and the exogenous component of the risk premium.

When $Y \leq Y^{ZLB}$, the extended *TR*-curve (99) implies that

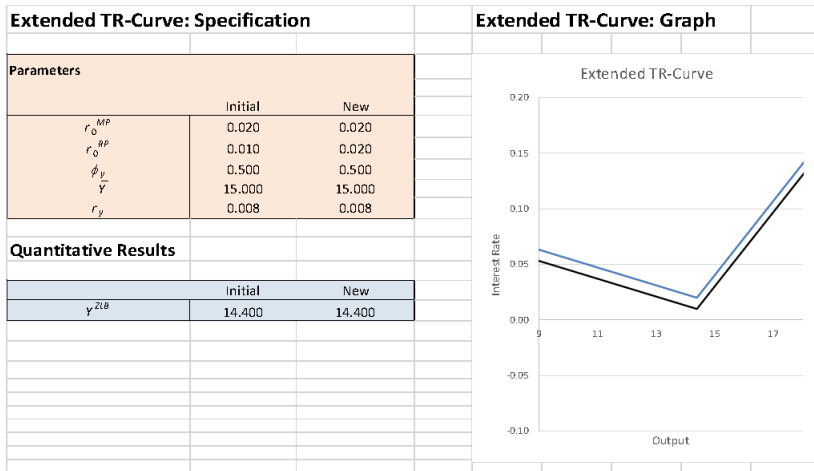
$$r = r_0^{RP} - r_y \cdot (Y - Y^{ZLB}). \quad (100)$$

The monetary policy rate then is constrained at the zero lower bound, and the interest rate at which households and firms can borrow is the sum of the exogenous and endogenous components of the risk premium.



Workbook for the Extended TR -Curve

The workbook *Extended-TR-Curve.xlsm* allows to study all the determinants of the position and slope of the extended *TR*-curve.



The extended *IS-TR* model combines the extended *TR*-curve with the standard *IS*-curve. Thus, the components of desired aggregate demand in the real sector in the extended *IS-TR* model are the same as in the *IS-TR* model, and are still given by:

$$C = C_0 + C_y \cdot (Y - T) - C_r \cdot r \quad (\text{see Equation (3)}),$$

$$I = I_0 - I_r \cdot r \quad (\text{see Equation (15)}),$$

$$G = G_0 + G_y \cdot (\bar{Y} - Y) \quad (\text{see Equation (21)}),$$

$$TB = TB_0 + TB_{ex} \cdot (Y^* - T^*) - TB_{im} \cdot (Y - T) + TB_{\varepsilon} \cdot \underbrace{[\varepsilon_0 - \varepsilon_r \cdot (r - r^*)]}_{= \varepsilon},$$

(see Equations (24) and (25)). Desired aggregate demand is thus still given by (49),

$$DD = A - (C_r + I_r + TB_{\varepsilon} \cdot \varepsilon_r) \cdot r + (C_y - TB_{im} - G_y) \cdot Y,$$

implying in turn that all combinations of output and the interest rate at which aggregate demand is equal to output supply are still represented by the *IS*-curve in (51),

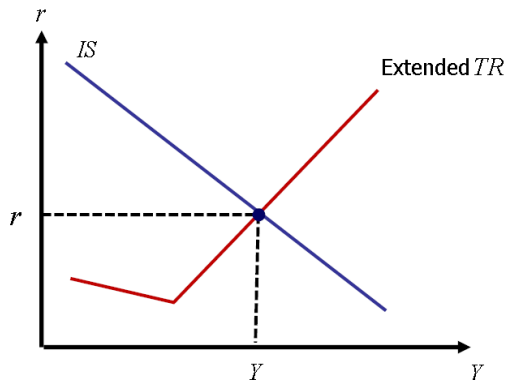
$$Y = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \frac{C_r + I_r + TB_{\varepsilon} \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot r.$$

The extended *IS-TR* model combines the standard real-sector *IS*-curve (51) with the financial-sector extended *TR*-curve (99),

$$r = \underbrace{\max \left\{ 0, r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) \right\}}_{=r^{MP}} + \underbrace{r_0^{RP} - r_y \cdot (Y - Y^{ZLB}) \cdot I}_{=r^{RP}} \cdot \mathbb{1}(Y \leq Y^{ZLB}).$$

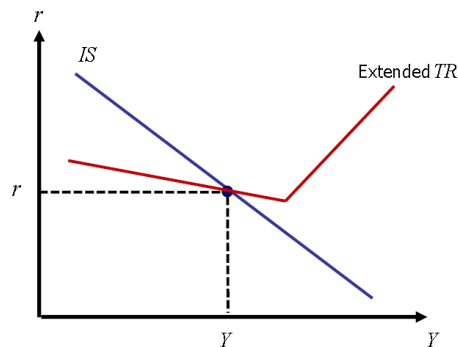
+ Short-run macroeconomic outcomes in the extended *IS-TR* model can thus be depicted ...

+ **"normal business cycle" outcome:**



"normal business cycle" outcome (along the upward-sloping portion of the extended *TR*-curve)

+ **"deep crisis" outcome:**



"deep crisis" outcome (along the downward-sloping portion of the extended *TR*-curve)

+ **Note:** The downward-sloping branch of the extended *TR*-curve from (100) has slope $-r_y$. Empirically, r_y is too small to warrant concerns that a scenario could arise in which the downward-sloping branch of the extended *TR*-curve does not intersect with the *IS*-curve.

Can the extended *IS-TR* model explain severe recessions, such as the one that followed the onset of the global financial crisis, and if so, what are its novel propagation channels relative to the *IS-TR* model?

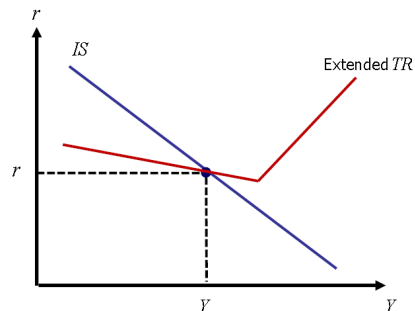
To address these questions, let us examine within the extended *IS-TR* model the short-run macroeconomic effects of the two shocks that seemed so important for the global financial crisis:

- a collapse in housing prices, implying $C_0 \searrow$, coupled with
- an increase in the exogenous component of the risk premium, $r_0^{RP} \nearrow$.

The decrease in housing prices and the associated decrease in C_0 cause the following adjustments:

- The *IS*-curve shifts to the left, the magnitude of the leftward shift determined by the size of the decrease in C_0 as well as the size of the Keynesian Multiplier.

Note: The shift from Point A to Point B is of the same magnitude as for the *IS*-curve in the *IS-TR* model, as it is for a given value of the interest rate.



"deep crisis" outcome (along the downward-sloping portion of the extended *TR*-curve)

- At Point B , we are off the extended TR -curve: As output has decreased from Point A to Point B , the monetary policy rate according to the central bank's Taylor rule is too high (the central bank aiming to stimulate macroeconomic activity).
- If there was no shock to the exogenous component of the risk premium, the macroeconomy would move from Point B to Point C . At Point C , simultaneous clearing of the real and financial sectors would be restored, with the central bank's easing of monetary policy reducing the recessionary impact of the decrease in C_0 (crowding in effects).
- However, given that there is also an increase of the exogenous component of the risk premium, r_0^{RP} (shifting the extended TR -curve up), the borrowing cost for households and firms is actually increasing, leading to crowding out effects.
- At Point D , the intersection of the new IS -curve with the new extended TR -curve, the interest rate is given by $r' > r$. The increase in the interest rate is also due to the endogenous component of the risk premium rising. The crowding out effects following the increase of r_0^{RP} cause output to fall below Y^{ZLB} , and the endogenous component of the risk premium then begins to become positive and rise, leading to further crowding out effects.
- As the interest rate at which households and firms can borrow increases,
 - households decrease consumption expenditure, reducing desired aggregate demand (crowding out through the consumption channel),
 - firms reduce investment in physical capital, also reducing desired aggregate demand (crowding out through the investment channel),
 - the domestic currency's spot value appreciates (financial investors moving their funds to the domestic economy, increasing the demand for domestic currency, as per the interest parity implication),
 - in turn, imports increase and exports decrease, that is, the trade balance decreases, reducing desired aggregate demand even further (crowding out through the exchange rate channel).

- The simultaneous new clearing of the real and financial sectors occurs at Point D – involving a lower level of output and a higher domestic interest rate.
- The overall decrease in output, Y , in the extended $IS-TR$ model is (potentially much) more severe than in the $IS-TR$ model, because
 - the central bank's attempts to stimulate monetary policy through lowering the monetary policy rate are constrained by the zero lower bound,
 - the risk premium can rise endogenously, further increasing the cost of borrowing for households and firms, and in turn further decreasing desired aggregate demand through the consumption, investment and exchange rate channels.

Let us combine the extended TR -curve with the IS -curve:

If $r^{MP} > 0$, (77) still holds:

$$\underbrace{\frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot r}{1 - (C_y - TB_{im} - G_y)}}_{\text{from } IS\text{-curve (51)}} = \underbrace{\bar{Y} + \left(\frac{\bar{Y}}{\phi_y} \right) \cdot (r - r_0^{MP} - r_0^{RP})}_{\text{from extended } TR\text{-curve when } r^{MP} > 0: (75)} = Y$$

If $r^{MP} = 0$, we have:

$$Y = \frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot (r_0^{RP} - r_y \cdot (Y - Y^{ZLB}))}{1 - (C_y - TB_{im} - G_y)} \quad (101)$$

from IS -curve (51) and extended TR -curve when $r^{MP} = 0$: (99)

Let us first turn to the case where $r^{MP} > 0$ ("**normal business cycle**" outcome):
The short-run rate of interest is then still given by (78):

$$r = \frac{\frac{A}{1 - (C_y - TB_{im} - G_y)} - \bar{Y} \cdot \left(1 - \frac{r_0^{MP} + r_0^{RP}}{\phi_y}\right)}{\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} + \frac{\bar{Y}}{\phi_y}}.$$

Having obtained the short-run rate of interest when $r^{MP} > 0$, we can also calculate short-run output from the *IS*-curve, (51),

$$Y = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot r.$$

We are then also in a position to plug back into our consumption, investment, government expenditure, trade balance and exchange rate functions to obtain the short-run outcomes for C , I , G , TB and ε for the case where $r^{MP} > 0$.

Turn second to the case where $r^{MP} = 0$ ("**deep crisis**" outcome): In this case, the short-run level of output can be obtained by solving (101) for Y :

$$Y = \frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot [r_0^{RP} - r_y \cdot (Y - Y^{ZLB})]}{1 - (C_y - TB_{im} - G_y)}$$

$$\Leftrightarrow \left[1 - \left(\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot r_y\right)\right] \cdot Y = \frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot (r_0^{RP} + r_y \cdot Y^{ZLB})}{1 - (C_y - TB_{im} - G_y)}$$

$$\Leftrightarrow Y = \frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot (r_0^{RP} + r_y \cdot Y^{ZLB})}{1 - (C_y - TB_{im} - G_y)} \Bigg/ \left(1 - \frac{(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot r_y}{1 - (C_y - TB_{im} - G_y)} \right), \quad (102)$$

where the level of Y^{ZLB} is given by (96):

$$Y^{ZLB} = \frac{\bar{Y}}{\phi_y} \cdot (\phi_y - r_0^{MP}).$$

The short-run rate of interest immediately follows from the extended TR -curve under $r^{MP} = 0$, Equation (100):

$$r = r_0^{RP} - r_y \cdot (Y - Y^{ZLB}), \quad (103)$$

with Y^{ZLB} again given by (96).

We are then once again in a position to also plug back into our consumption, investment, government expenditure, trade balance and exchange rate functions to obtain the short-run outcomes for C, I, G, TB and ε for the case where $r^{MP} = 0$.

Numerical Example:

Suppose $A = 9.78$, $C_y = 0.6$, $G_y = 0.15$, $TB_{im} = 0.05$, $TB_{ex} = 0.01$,
 $C_r = 4$, $I_r = 16$, $\bar{Y} = 15$, $r_0^{MP} = 0.02$, $r_0^{RP} = 0.01$, $\phi_y = 0.5$,
 $r_y = 0.005$, $TB_\varepsilon = 1$, $\varepsilon_r = 6$.

A new complication arising now is that (in general) we do not know prior to our calculations whether the short-run outcome is a "normal business cycle" outcome or a "deep crisis" outcome.

One approach to resolving this complication is as follows:

Step 1:

Conjecture that we are in a "deep crisis" outcome. (We make this conjecture as the algebra to solve the model is a bit less involved for the "deep crisis" outcome than for the "normal business cycle" outcome).

Based on this conjecture, calculate the implied level of output from Equations (102) and (96):

$$Y_{Conjecture} = \frac{A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot (r_0^{RP} + r_y \cdot Y^{ZLB})}{1 - (C_y - TB_{im} - G_y)} \bigg/ \left(1 - \frac{(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot r_y}{1 - (C_y - TB_{im} - G_y)} \right),$$

where

$$Y^{ZLB} = \frac{\bar{Y}}{\phi_y} \cdot (\phi_y - r_0^{MP}).$$

Step 2:

If the conjectured level of output calculated in Step 1 was less than or equal to Y^{ZLB} ,

$$Y_{Conjecture} \leq Y^{ZLB},$$

then indeed we are in the "deep crisis" outcome, for which it then must also hold that r^{MP} is constrained to be at the zero lower bound, that is, from the Taylor rule in Equation (62) we have that

$$r^{MP} = r_0^{MP} + \phi_y \cdot \left(\frac{Y_{Conjecture} - \bar{Y}}{\bar{Y}} \right) \leq 0.$$

We can then calculate the outcomes for all variables based on the relations that hold for the "deep crisis" case, including in particular that

$$Y = Y_{Conjecture}.$$

Step 3:

If, however, the conjectured level of output calculated in Step 1 was larger than Y^{ZLB} ,

$$Y_{Conjecture} > Y^{ZLB},$$

then a contradiction has arisen, and the outcome must be a "normal business cycle" outcome, so that we need to calculate all outcomes based on the relations that hold when $r^{MP} > 0$: In particular, we can need to calculate short-run output from the *IS*-curve, (51),

$$Y = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot r,$$

with the short-run interest rate, r , from Equation (78) given by

$$r = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \bar{Y} \cdot \left(1 - \frac{r_0^{MP} + r_0^{RP}}{\phi_y} \right) \bigg/ \left(\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} + \frac{\bar{Y}}{\phi_y} \right).$$

Implementing this procedure for our numerical example:

- + Presume at first that we are in the "deep crisis" outcome, where $r^{MP} = 0$:
From Equations (96) and (102) we obtain $Y_{Conjecture} \approx 16.272$. However, $Y^{ZLB} = 14.4$ and the Taylor rule (see Equation (62)) for $Y \approx 16.272$ would imply that $r^{MP} \approx 0.062$, both contradictions to us being in a "deep crisis" outcome.
- + We then must be in the "normal business cycle" outcome, where $r^{MP} > 0$:
Using our parameter values in Equation (78), we obtain $r = 0.03$. Using this result in the *IS*-relationship, we obtain $Y = 15$. This in turn from the Taylor rule implies that $r^{MP} = 0.02 = r - r_0^{RP}$.

Now suppose $\Delta C_0 = -0.7$ and thus $A' = 9.08$, and that $\Delta r_0^{RP} = 0.04$ and thus $(r_0^{RP})' = 0.05$.

- Presume at first that we are in the "deep crisis" outcome, so that $(r^{MP})' = 0$:
From Equations (96) and (102) we now obtain $Y^{Conjecture} \approx 12.570$ and $Y^{ZLB} = 14.4$. Also, for $Y' \approx 12.570$, the Taylor rule would imply that $(r^{MP})' \approx -0.061$.
So we are indeed in the "deep crisis" outcome. From Equation (103), we obtain $r' \approx 0.059$.

Let us decompose the overall change in output that has occurred:

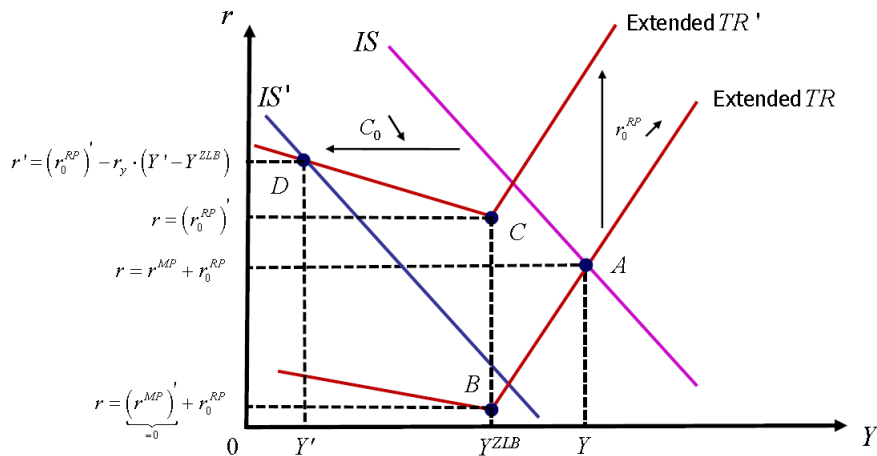
- Starting point for such a decomposition is again the *IS*-curve, Equation (51):

$$Y = \frac{A}{1 - (C_y - TB_{im} - G_y)} - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot r.$$

- Following the change in both C_0 (and thus A) as well as r_0^{RP} , the standard comparative statics analysis (see, for example, Equation (81)) would seem to apply:

$$\Delta Y = \frac{\partial Y}{\partial A} \cdot \Delta A + \frac{\partial Y}{\partial r} \cdot \Delta r.$$

- Note, however, that before the change in C_0 and r_0^{RP} , it held that $Y > Y^{ZLB}$, and that after the change in C_0 and r_0^{RP} , it holds that $Y' < Y^{ZLB}$.
- The short-run macroeconomic outcome thus has switched from a "normal business cycle" outcome (along the upward-sloping portion of the extended *TR*-curve) to a "deep crisis" outcome (along the downward-sloping portion of the extended *TR*-curve).
- As the extended *TR*-curve is not differentiable at Y^{ZLB} , the overall change in the interest rate, Δr , can thus not be obtained using differentiation-based techniques only.
- The following graphical analysis-based decomposition provides one means to calculate Δr :



What is Δr ?

$$\Delta r \Big|_{\text{Point D from Point A}} = \underbrace{-r^{MP}}_{\Delta r \Big|_{\text{Point B from Point A}}} + \underbrace{(r_0^{RP})' - r_0^{RP}}_{\Delta r \Big|_{\text{Point C from Point B}}} - \underbrace{r_y \cdot (Y' - Y^{ZLB})}_{\Delta r \Big|_{\text{Point D from Point C}}}.$$

- From Point A to Point B: The monetary policy rate has decreased from $r^{MP} > 0$ to $(r^{MP})' = 0$. The associated change in the interest rate is

$$\Delta r|_{\text{Point B from Point A}} = r|_{\text{Point B}} - r|_{\text{Point A}} = \underbrace{r_0^{RP}}_{\text{Point B}} - \underbrace{(r^{MP} + r_0^{RP})}_{\text{Point A}} = -r^{MP}. \quad (104)$$

- From Point B to Point C: The exogenous component of the risk premium has increased from r_0^{RP} to $(r_0^{RP})'$. The associated change in the interest rate is

$$\Delta r|_{\text{Point C from Point B}} = r|_{\text{Point C}} - r|_{\text{Point B}} = \underbrace{(r_0^{RP})'}_{\text{Point C}} - \underbrace{r_0^{RP}}_{\text{Point B}} = \Delta r_0^{RP}. \quad (105)$$

- From Point C to Point D: The endogenous component of the risk premium has increased from 0 to $-r_y \cdot (Y' - Y^{ZLB})$. The associated change in the interest rate is

$$\Delta r|_{\text{Point D from Point C}} = r|_{\text{Point D}} - r|_{\text{Point C}} = \underbrace{(r_0^{RP})' - r_y \cdot (Y' - Y^{ZLB})}_{\text{Point D}} - \underbrace{(r_0^{RP})'}_{\text{Point C}} = -r_y \cdot (Y' - Y^{ZLB}). \quad (106)$$

Equations (81), (104), (105) and (106) give us the following decomposition of the overall change in output:

- Change in output due to Keynesian Multiplier effect:

$$\frac{\partial Y}{\partial A} \cdot \Delta A = \frac{\Delta A}{1 - (C_y - TB_{im} - G_y)} \approx -1.167.$$

Equations (81), (104), (105) and (106) give us the following decomposition of the overall change in output:

- Change in output due to crowding in of consumption, investment and the trade balance caused by the decrease of the monetary policy rate to zero:

$$\frac{\partial Y}{\partial r} \cdot (-r^{MP}) = -\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot (-r^{MP}) \approx 0.867.$$

- Change in output due to crowding out of consumption, investment and the trade balance caused by the increase of the exogenous component of the risk premium:

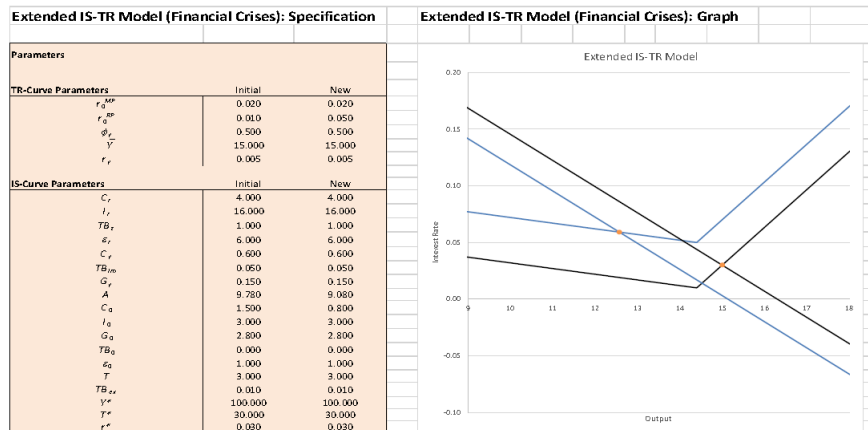
$$\frac{\partial Y}{\partial r} \cdot \Delta r_0^{RP} = -\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r_0^{RP} \approx -1.733.$$

- Change in output due to crowding out of consumption, investment and the trade balance caused by the increase of the endogenous component of the risk premium:

$$\frac{\partial Y}{\partial r} \cdot (-r_y) \cdot (Y' - Y^{ZLB}) = -\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)} \cdot (-r_y) \cdot (Y' - Y^{ZLB}) \approx -0.396.$$

Workbook for the Extended IS-TR Model (Financial Crises)

The workbook *Extended-IS-TR-Model-FinancialCrises.xlsm* allows to study how changes in the autonomous components of aggregate demand and/or other model parameters affect output in the ("deep crisis" or "normal business cycle") short-run macroeconomic outcome, ...



... how these output effects may be decomposed, and how the other endogenous variables change:

Extended IS-TR Model (Financial Crises): Keynesian Multiplier and Consumption, Investment as well as Exchange Rate Channels: Quantitative Results					Initial and New Short-Run Macroeconomic Outcomes			
Transmission Channels for Changes in Output Following Changes Exclusively in the Components of Autonomous Demand and/or Exogenous Component of Risk Premium					Initial and New Short-Run Macroeconomic Outcomes Following Changes in Any of the Model Parameters			
Keynesian Multiplier					Initial	New	Absolute Change	Percentage Change
Stimulus: Change in A				-0.700	15.000	12.570	-2.430	-16.199
Size of Multiplier				1.667	0.030	0.059	0.029	97.163
Change in Output				-1.167	0.020	0.000	-0.020	-100.000
Interest Rate Implied Changes in Output					8.580	6.306	-2.274	-26.509
					2.520	2.054	-0.466	-18.507
					2.800	3.164	0.364	13.017
					1.100	1.047	-0.053	-4.855
					1.000	0.825	-0.175	-17.489
Interest Rate Change: Consumption Channel								
Change in Output	r^{RP}	r_0^{RP}	endog. comp. of r^{RP}	r				
Interest Rate Change: Investment Channel								
Change in Output								
Interest Rate Change: Exchange Rate Channel								
Change in Output								
Sum of Interest Rate Implied Changes in Output								
Change in Output								
Total Change in Output								
Sum of Effects								